

Semantics of Comparative Constructions

Class 2: More on adjectives

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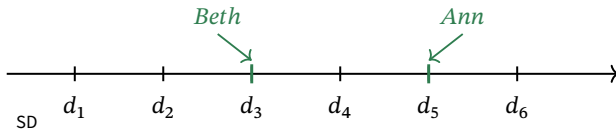
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Part 1

Recap & today's plans

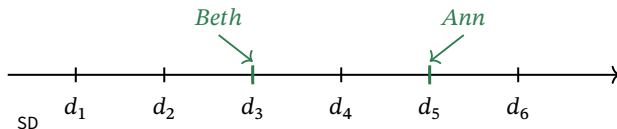
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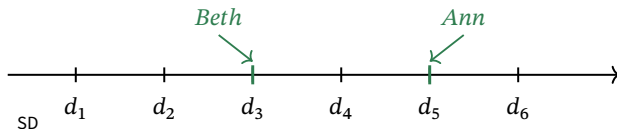


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- We'll use degrees to analyze gradable predicates as relation individuals & degrees:
 - (1) x is d -tall $\approx x$ is tall to degree d
- We added a new basic type d .

Today

- We'll cover the following topics related to the interpretation of adjectives:
 - * the basic meaning of adjectives
 - * vagueness & the positive form
 - * degree modification & scale structure
 - * antonyms

Part 2

Interpreting adjectives

Adjectives with measure phrases

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(said while showing distance with hands)

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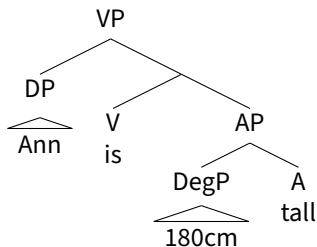
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- We assume that these sentences have the following underlying structure:

(3)



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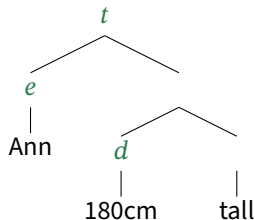
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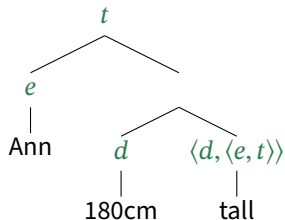
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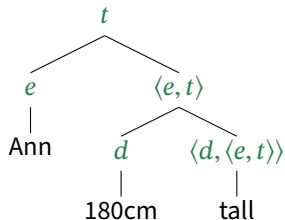
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‘Exact’ semantics for adjectives

- A simple analysis of *tall* would be to map an individual to its exact height:

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 - a relation between the degree argument and the output of the measure function
- A potential challenge:

(7) You have to be 180cm tall to ride the roller coaster.

$\rightsquigarrow$ *you have to be **at least** 180 tall*

‘At least’ semantics for adjectives

- We thus give adjectives an ‘at least’ semantics:

$$(8) \quad \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{height}(x) \geq_{\text{tall}} d \qquad \qquad \qquad :: \langle d, \langle e, t \rangle \rangle$$

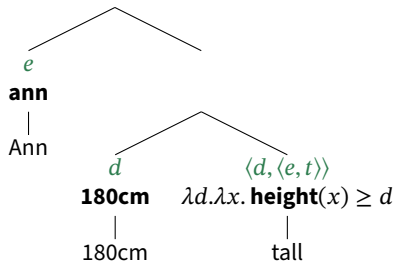
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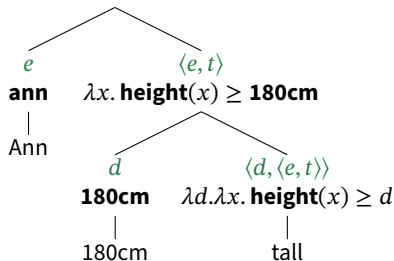
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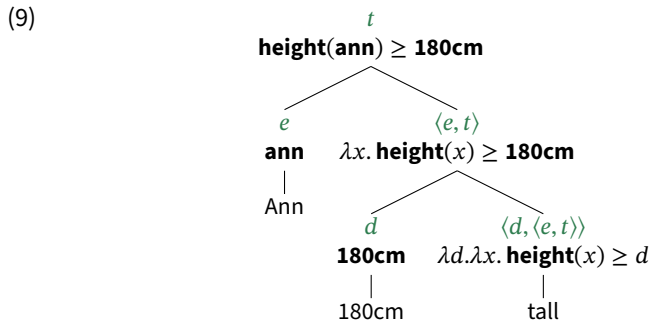


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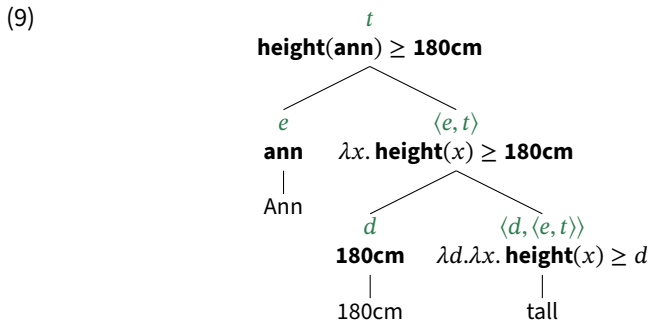


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- Putting it all together:



- The meaning we get, then, is that *Ann is 180cm tall* is true if Ann's height is *at least* 180cm.
- We assume the 'exact' interpretation arises from pragmatic strengthening.

Part 3

On the semantics of the positive

The positive form of adjectives

- Adjectives their bare form–the **positive**–leads to vagueness and context variability:

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- But in a way, that's by design! $\rightsquigarrow$ *vagueness disappears in (**some**) other forms of adjectives!*

- (12) Paolo is 1mm **taller** than my knees.
(13) I know **how tall** my friend Ann is: exactly 153cm.
(14) Ann is **150cm** tall.

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- We'll assume that source of vagueness and context variability is a null morpheme POS:

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- Entry for POS:

(17) $\llbracket \text{POS} \rrbracket = \lambda A_{\langle d, \langle e, t \rangle \rangle}. \lambda P_{\langle e, t \rangle}. \lambda x_e. \exists d. A(d)(x) \wedge d \text{ !> } \text{stnd}(A)(P)$

Part 4

Degree modifiers & scale structure

Degree modifiers

- Gradable adjectives can often be combined with **degree modifiers**:

- (18)
- a. Her eyes were **half** open.
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- However, these modifiers show sensitivity to the kind of adjective they can combine with:

- (19)
- a. #My pet snake is **half** old.
 - b. #My new bike was **mostly** expensive.
 - c. #This area is **completely** dangerous.

(see Bolinger 1972, Lehrer 1985, Cruse 1986, and references in the following slides)

Scales & endpoints

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- Adjectives differ in whether the scales associated with them have endpoints:

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b. x is completely *tall* $\rightsquigarrow \#x$ **is tall to the maximal degree one can be tall**

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- Our degree-based semantics allows us to model these ideas precisely.

Four types of scales

- Let us exploit again the fact that scales and intervals of real numbers are isomorphic.
- Consider the two intervals in (22), which differ in their upper edge:
 - * interval (22a) has a **closed** upper endpoint: it's maximal element is 1
 - * interval (22a) has a **open** upper endpoint: it has no maximal element

(22) a. $\{r \in \mathbb{R} \mid 0 \leq r \leq 1\}$

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* **(totally) closed:** e.g. *full*

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- Simplified entries for *half* and *fully*:

- (25) a. x is half $A \rightsquigarrow \exists d. x$ is A to $d \wedge$ the difference between $\mathbf{max}(S_A)$ and d
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b. $x \text{ is fully } A \rightsquigarrow x \text{ is } A \text{ to } \mathbf{max}(S_A)$

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 - adjectives with at least one closed end don't ⇒ **absolute adjectives**

(see Kennedy and McNally 2005, Kennedy 2007 and also Rotstein and Winter 2004 and references therein)

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(26) **Relative adjectives**

a. Ann is old.

↪ *unclear boundaries*

b. Ann is young.

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(26) **Relative adjectives**

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b. Ann is young.

↪ *unclear boundaries*

(27) **Absolute predicates**

a. The rod is straight.

↪ *true if the rod is fully straight*

b. The rod is bent.

↪ *true if the rod isn't fully straight*

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- adjectives with at least one closed end don't

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(see Kennedy and McNally 2005, Kennedy 2007 and also Rotstein and Winter 2004 and references therein)

(26) **Relative adjectives**

a. Ann is old.

↪ *unclear boundaries*

b. Ann is young.

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(27) **Absolute predicates**

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- **Kennedy 2007:** transitions to and from maximal endpoints 'naturally' stand out
- **McNally 2011:** some counterexamples
 - * *familiar* has a closed scale (*completely familiar*), but the standard is contextually set
 - * the standard for *full* seems sensitive to context, e.g., a glass of wine

Part 5

Antonyms & measure phrases

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bent _{lower closed}	<i>vs</i>	straight _{upper closed}
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full _{closed}	<i>VS</i>	empty _{closed}

- Furthermore, they usually often license inferences such as the following:

- (28) a. Ann is taller than Beth.
 $\Leftrightarrow$ Beth is shorter than Ann.
- b. The stork's beak is longer than the sparrow's.
 $\Leftrightarrow$ The sparrow's beak is shorter than the stork's.

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 - b. six feet wide/#narrow
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⇒ We'll now focus on the first of these.

Same degrees, different relation

- A typical analysis of antonym pairs α^+, α^- :
 - if α^+ is associated with a scale $\langle S, \leq_S \rangle$,
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- Our analysis of measure phrases, however, predicts no contrast:

$$(34) \quad \begin{array}{ll} \text{a.} & \text{Ann is 180cm tall.} \quad \rightsquigarrow \mathbf{height(ann)} \geq \mathbf{180cm} \\ \text{b.} & * \text{Ann is 180cm short.} \quad \rightsquigarrow \mathbf{180cm} \geq \mathbf{height(ann)} \end{array}$$

Intervals

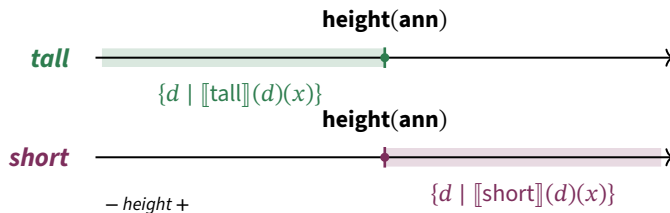
- Antonyms relate the same individuals to different degrees on a scale.
- We can use them to identify different **intervals**:

(35) a. $\{d \mid \llbracket \text{tall} \rrbracket(d)(\mathbf{ann})\} = \{d \mid \mathbf{height}(\mathbf{ann}) \geq d\}$
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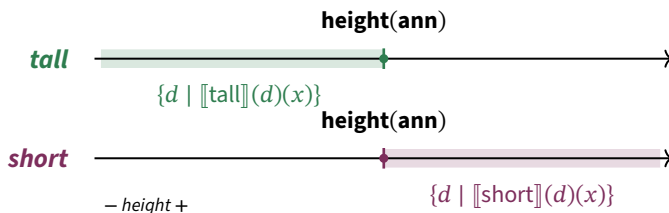
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⇒ Only one of these intervals could be said to measure 180cm! (or anything at all)

Measure phrases as measuring intervals

- With this in mind, we could say that, instead of denoting degree names, measure phrases are functions that measure scale segments: (Schwarzschild 2005)

$$(36) \quad \llbracket 180\text{cm} \rrbracket = \lambda D_{dt}. \mathbf{180cm}(D)$$

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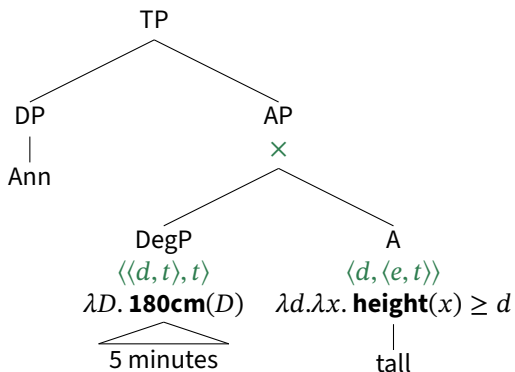
- The set created from *short*, however, is unbounded, and thus can't be measured:

$$(38) \quad (\lambda d. d \geq \mathbf{height(ann)}) = [\mathbf{height(ann)}, \infty)$$

(this kind of analysis has a lot of precedent: Hale 1970, Bierwisch 1989, Seuren 1978, Faller 2000, Schwarzschild 2005)

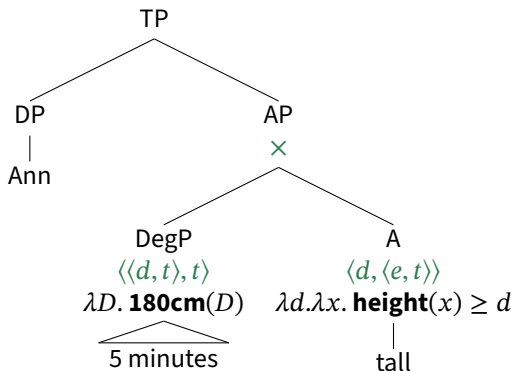
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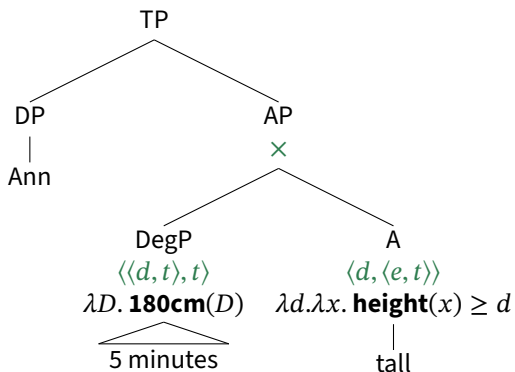
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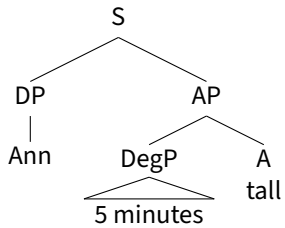
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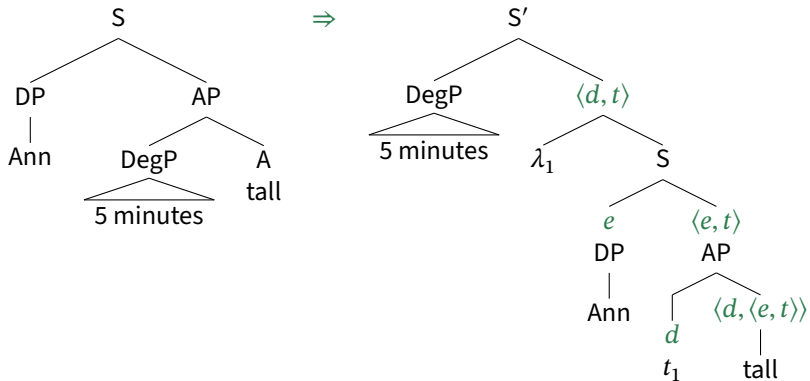


- Furthermore, how do we get access to intervals compositionally?
- Hint:** what does the type of DegP remind you of?

QR of DegP, part I

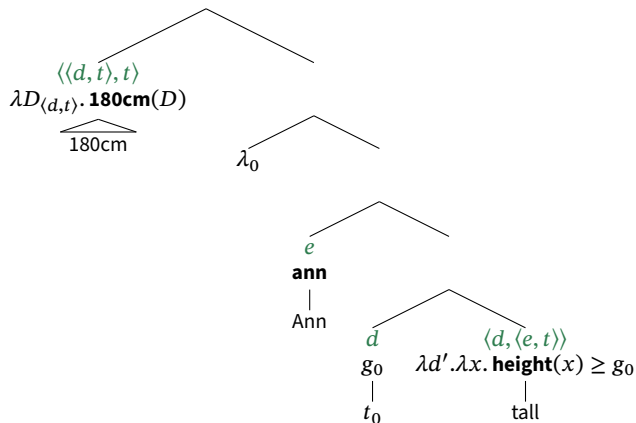


QR of DegP, part I



QR of DegP, part II

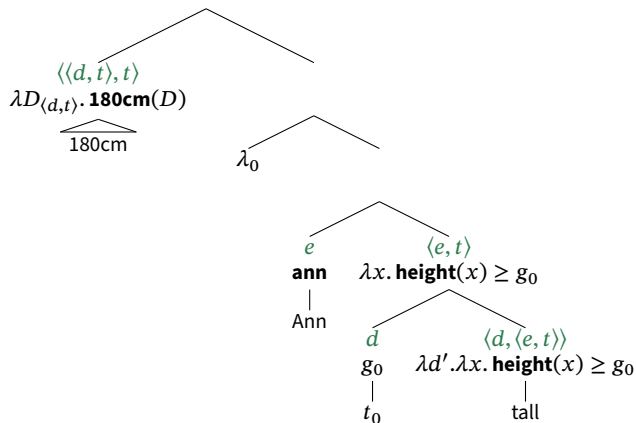
(39)



⇒ **Lesson:** Movement of DegP create an interval denoting constituent.

QR of DegP, part II

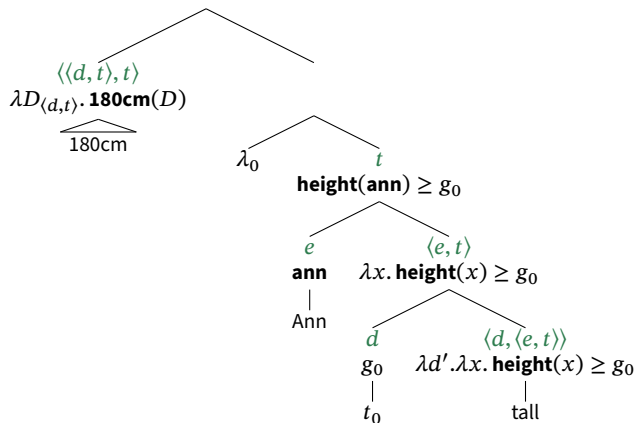
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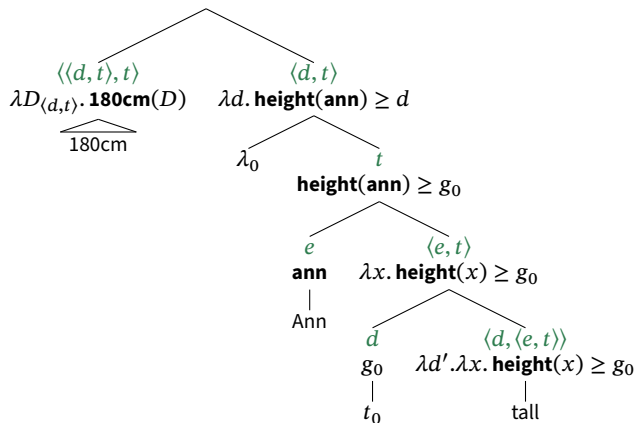
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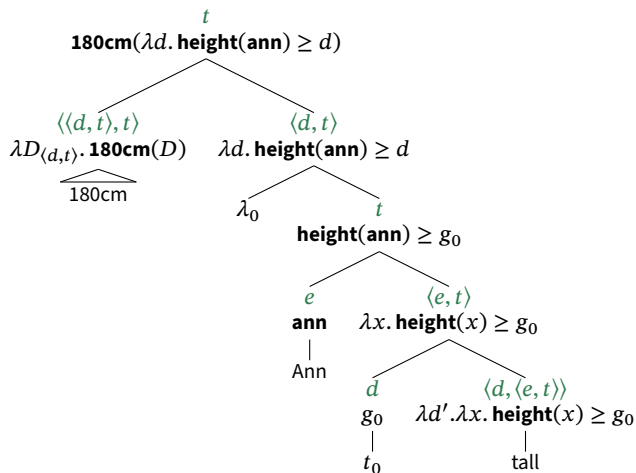
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- a. *The bag is 20kg heavy/light.
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- For further discussion on these matters, see Faller 2000, Winter 2005, Sassoön 2010a, a.o.

Part 6

Summing things up

Summary (or: what do remember for tomorrow)

- Adjectives have relate individuals to degrees via measure functions and the $\geq$ -relation:

$$(43) \quad \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{tall}(x) \geq d$$

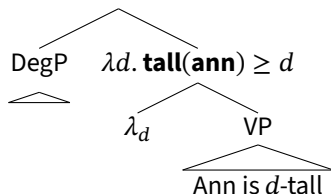
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- At least meanings give us access to **intervals**.

$$(44) \quad \text{Ann is [DegP tall]} \Rightarrow$$



⇒ DegP movement creates interval-denoting constituents

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Ambiguities with measure phrases

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- (45) The paper is required to be 5 pages long.
- a. **Exactly**: in every world w where the rules are obeyed, the paper's length $w = 5pp$
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(46) a. required $[180cm \lambda_d [the\ paper\ is\ d\text{-}long]]$

- b. $\forall w \in \mathbf{Rule}(w_0). \mathbf{180cm}(\lambda d. \mathbf{length(the.paper)}(w) \geq d)$

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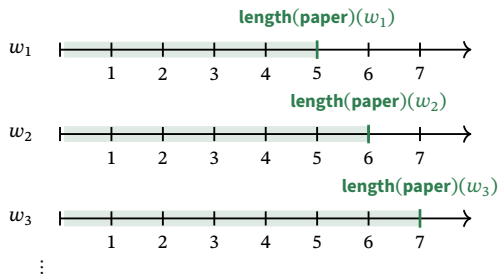
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- But what happens when the measure phrase scopes over *require*?

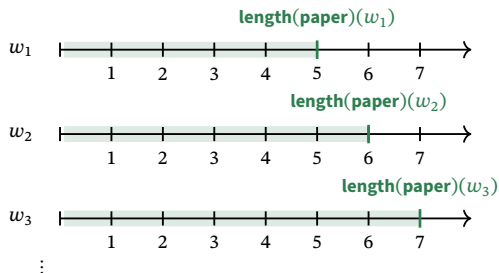
DegP > require

Worlds where the rules are obeyed:



DegP > *require*

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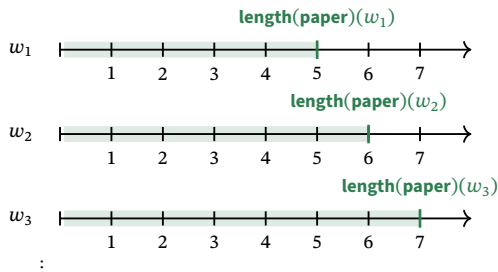


- Suppose DegP moves over *require*:

(47) 180cm λ_d [required [the paper is t_d -long]]

DegP > require

Worlds where the rules are obeyed:



- Suppose DegP moves over *require*:

(47) $180\text{cm } \lambda_d [\text{required} [\text{the paper is } t_d\text{-long}]]$

- The sister of DegP denotes the following interval:

(48) $\lambda d. \forall w \in \mathbf{Rule}(w_0) [\text{length}(\text{paper})(w) \geq d]$

$\rightsquigarrow$ this function is true of which degrees?